Self-Driving Lab Simulator Using Q-Learning

Optimizing Chemical Reaction Yield through Reinforcement Learning Parameter Search

Manual Bottleneck

Traditional trial-and-error parameter tuning in labs is slow, costly, and resource-intensive.

Autonomous AI

RL agents learn high-yielding experimental conditions automatically through closed-loop interaction.

Project Goal

Maximize simulated reaction yield by autonomously adjusting temperature & concentration.

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Problem Formulation & Algorithm

STATE SPACE (S)

121 States

11 Temp bins (30–100°C)
11 Conc bins (0.1–1.0 rel)

ACTION SPACE (A)

9 Actions

Local step choices
ΔT, ΔC ∈ {-1, 0, +1}

REWARD FUNCTION

Yield / 100

Normalized reaction output in range [0, 1] + noise

HYPERPARAMETERS

Q-Table Setup

α = 0.18, γ = 0.92
ε decay: 1.0 → 0.04

Update Equation: $Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$

Algorithm: Tabular Q-Learning (Model-Free)

SLIDE 3 — RESULTS

Learning Curves & Comparison

Experimental Outcomes

- **700 Episodes Trained:** Executed 35 simulation steps per training episode.
- **Q-Agent Growth:** Yield improved from ~15% to **>80%** mean reaction yield.
- **Random Baseline:** Stayed flat around **~15% yield** due to unguided exploration.
- **Optimal Point Found:** Successfully localized target peak near **74°C** and **0.68 conc**.

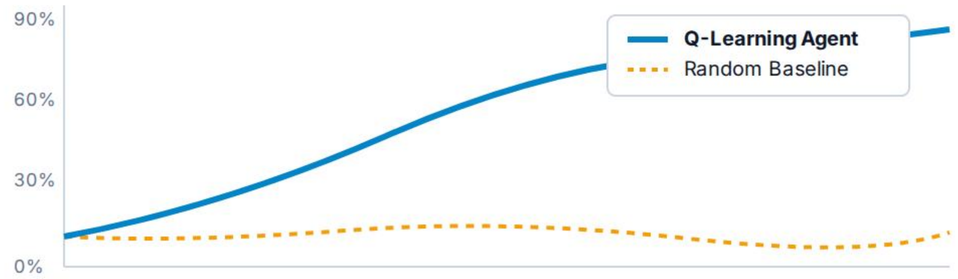


Figure 1: Mean Reaction Yield per Episode (30-ep moving average)

Evaluated across 700 episodes in Matplotlib

Simulator Constraints

Synthetic Model

Built on a simplified mathematical surface; does not reflect real chemical reaction kinetics or complex multi-phase dynamics.

Discrete Bins

Tabular Q-learning restricts parameter optimization to 121 discrete grid points rather than continuous control spaces.

Physical Constraints

Ignores real-world lab limits such as thermal ramp times, chemical costs, reagent consumption, and equipment strain.

Future Enhancements: Continuous Deep RL (PPO/DQN) & Bayesian Optimization

Responsible Use & Project Summary

Ethical Considerations

- **Lab Safety Validation:** Autonomous recommendations must undergo strict safety checks before real chemical execution.
- **Human Oversight:** AI serves as decision support; expert chemists must supervise automated systems.
- **Responsible Use:** Ensure automated lab algorithms cannot be repurposed for hazardous synthesis.

Final Conclusion

The Self-Driving Lab Simulator demonstrates that Q-learning effectively automates parameter optimization in a closed-loop setting. The agent successfully learned high-yielding temperature and concentration settings, serving as a foundational baseline for intelligent lab automation.